Fitness Athlete Club (Titan Iron Gym)

Object Oriented Analysis and Software Engineering (OOASE) – Project Report

Salem, Tamil Nadu • Live Web Application Deployment

1. Introduction

1.1 Overview

Fitness Athlete Club (operating as Titan Iron Gym) is an advanced, high-performance web-based athletic club and gymnasium management system specifically engineered for Olympic powerlifting facilities, strength academies, and commercial gymnasiums in Salem, Tamil Nadu. The system centralizes member registration, athletic progress tracking, fast touchless QR/barcode attendance scanning, personal coach assignments, Indian Rupee (INR ₹) tier billing, and a real-time e-commerce supplement store.

The software guarantees robust role-based security by strictly separating normal customer/member portals from the restricted Creator & Master Admin portal. While public users log in to their personalized Member Profile to upload profile pictures (DP), update body metrics (height, weight, body fat percentage), and accept club safety liability waivers, the website creator exercises full administrative control over member rosters, coach assignments, financial revenue reports, and facility operations through a discreet master key portal.

The project was designed, modeled, and developed following strict Object-Oriented Analysis and Software Engineering (OOASE) principles: requirements specifications, comprehensive UML modeling (Use Case, Class, Sequence, Activity, State Chart, Component, and Deployment diagrams), client-side single-page architecture (SPA) in modern JavaScript (ES6+), persistent browser LocalStorage data modeling, responsive Vanilla CSS3 dark-mode aesthetics, and production deployment on Vercel's global edge network.

1.2 Problem Statement

Conventional athletic clubs and private gyms often rely on manual paper logbooks, isolated spreadsheets, and unmonitored counter transactions, leading to critical operational bottlenecks:

- **Unrestricted Entry & Expired Memberships:** Lack of real-time check-in validation allows expired or unauthorized members onto the training floor.
- **Scattered Athlete Data:** Personal contact info, health waivers, height/weight logs, and training goals are kept across loose paper forms.
- **Exposed Administrative Controls:** Traditional gym sites display master admin login inputs on the public login page, risking credential theft by regular visitors.
- **Absence of Digital Identity:** Members lack digital photo identification and touchless check-in passes.
- **Disconnected Nutrition Sales:** Protein powders and gym supplements are sold manually without inventory tracking, shopping cart calculation, or instant UPI payment verification.

- **Lack of Real-Time Reporting:** Attendance rates, expiring memberships, and daily collection figures require hours of manual calculation.

1.3 Objectives

- Centralize athletic member directories, certified coaches, membership tiers, and pro shop products in an integrated web platform.
- Provide a dedicated **Customer Profile & Fitness Dashboard** where athletes can upload profile pictures (DP) that stay permanently saved, log fitness body metrics, track workout streaks, and agree to club health waivers.
- Implement a **Restricted Creator Admin Master Key** accessed solely via discreet golden key triggers, completely shielded from standard member login pages.
- Enable real-time **Touchless Barcode / QR Code Check-in** with instantaneous access grant or expiry denial.
- Deliver an authentic **Pro Supplement Shop** with real-time cart drawer, coupon application, and Google Pay (GPay) / PhonePe UPI payment simulation.
- Model the complete system architecture using standard object-oriented UML diagrams.
- Deploy the production application globally with automated CI/CD on Vercel: `https://fitness-athlete-club.vercel.app/`.

1.4 Scope of the Project

- **Included:** Member registration and authentication, customer profile dashboard, permanent DP photo upload, fitness body metrics tracking, club safety and medical waiver acceptance, member directory with search and status filters, printable digital ID passes, 1-click renewal with automated invoice generation, certified trainer profiles (featuring leading Tamil fitness athletes such as Surabi, Dolphin, Biglee Murali, Saran Fitness, and Kabilan), pro shop with real-time cart, UPI QR code generation, CSV data export, and dark-mode athletic user interface.
- **Not included:** Native mobile app stores (built as a mobile-first progressive web responsive app), hardware biometric turnstiles, and bank credit card merchant accounts requiring physical POS swipe hardware.

1.5 Technologies Used

Layer / Function	Technology	Purpose
Frontend UI	HTML5, Modern CSS3	Responsive single-page layout, athletic dark mode (Obsidian Black & Gym Red), Google Fonts (Inter & Oswald).
Logic & Router	Vanilla JavaScript (ES6+)	Single Page Application (SPA) view router, form validation, modal controls, real-time clock, and DOM controller.
State & Storage	HTML5 LocalStorage Engine (GymStore)	Persistent client-side data store with versioned schema seeding, instant CRUD operations, and permanent DP image persistence.

Layer / Function	Technology	Purpose
Security & Auth	Role-Based Access Control (RBAC)	Separation of Customer/Member sessions from Creator Master Key access; discreet master modal unlock.
File Reader	HTML5 FileReader API	Client-side binary-to-Base64 image encoding with 3MB file limits for permanent DP uploads.
E-Commerce & UPI	JavaScript Cart & QRServer API	Dynamic price calculation, cart drawer, discount coupon parsing, and dynamic UPI QR code rendering for GPay / PhonePe.
Version Control	Git, GitHub	Source code repository, branch tracking, and commit history (`mohammedyusuf-a/fitness-athlete-club`).
Hosting & CI/CD	Vercel Edge Platform	High-speed global CDN deployment with automated push-to-deploy pipeline.
OOASE Modeling	UML (Unified Modeling Language)	Visual software architecture models: Use Case, Class, Sequence, Activity, State Chart, Component, and Deployment.

1.6 Organisation of the Report

Section 2 details the requirement analysis, actor definitions, functional and non-functional requirements, data dictionary, and business rules. Section 3 presents the seven standard UML diagrams with comprehensive design descriptions. Section 4 covers the software coding, directory structure, data models, and testing scenarios. Section 5 presents full application output screenshots. Section 6 concludes the report with limitations, future enhancements, and academic references.

2. Requirement Analysis

2.1 Approach

Requirements were gathered by analyzing real-world gymnasium workflows at Olympic athletic facilities in Tamil Nadu. The object-oriented approach was adopted: primary system actors were identified, their functional goals decomposed into discrete use cases, domain entities mapped to class hierarchies, and operational constraints translated into enforceable business rules.

2.2 User Roles

Role	Description	Main Responsibilities
Website Creator & Master Admin	Sole system owner possessing absolute administrative control.	Accesses the facility via discreet master key; manages member accounts, registers new coaches, edits membership fees, inspects billing ledgers, exports CSV rosters, and resets demo datasets.
Gym Member / Athlete	Registered club subscriber (e.g. Md Yusuf).	Signs in to Member Portal; uploads and updates permanent profile picture (DP); manages fitness metrics (height, weight, body fat); views fast check-in QR pass; agrees to club terms & waivers; shops for protein powders; logs out.
Certified Trainer / Coach	Athletic trainers (e.g. Saran, Biglee, Kabilan).	Supervises assigned client athletes, reviews specialized training targets (Powerlifting, Hypertrophy, Turf Conditioning), and guides fitness objectives.
Guest Athlete	Prospective athlete or non-registered visitor.	Explores club training facilities, reviews membership tier pricing, browses authentic supplement inventory, and registers a new account.

2.3 Functional Requirements

ID	Requirement Description	Module
FR-01	Allow member sign-in and new account registration with input validation.	Authentication
FR-02	Provide a clean public login page without exposing administrative portal credentials.	Security
FR-03	Provide dedicated, discreet Creator Admin master key triggers (floating key, topbar, sidebar) leading to restricted PIN authorization.	Admin Portal

ID	Requirement Description	Module
FR-04	Provide a dedicated Member Profile Dashboard for logged-in athletes.	Member Portal
FR-05	Allow members to upload a profile picture (DP) using FileReader Base64, persisting it permanently across reloads and sessions.	Profile & Media
FR-06	Allow members to update personal details (Full Name, Email, Phone, DOB, Emergency Contact, Address).	Profile Management
FR-07	Allow members to upload and track athletic body metrics (Height, Weight, Target Weight, Body Fat %, Goal, Coach).	Fitness Metrics
FR-08	Present an interactive Club Terms, Gym Safety Rules & Medical Waiver scroll box with agreement checkbox.	Legal & Compliance
FR-09	Display a touchless Digital Fast Check-in QR pass on the member profile card.	Check-in / Identity
FR-10	Provide single-click member logout returning cleanly to the public login gateway.	Session Control
FR-11	Display executive dashboard KPIs (Active Members, Today's Check-ins, Expiring Members, Monthly Revenue).	Executive Dashboard
FR-12	Provide a header-mounted barcode/ID scanner for instant 1-click check-in and checkout.	Attendance
FR-13	Maintain searchable members directory with plan and status filters.	Members Directory
FR-14	Display printable Digital Athletic ID cards with photo, barcode, and emergency contact.	Digital ID Badge
FR-15	Enable 1-click membership renewal with automated invoice and receipt generation.	Billing & Renewal
FR-16	Support real-time Pro Supplement Shop with category filtering and interactive quick-view modal.	Pro Shop
FR-17	Maintain a persistent real-time shopping cart with discount coupon support (e.g. TITAN20).	Shopping Cart
FR-18	Render dynamic UPI QR codes for instant Google Pay (GPay) and PhonePe checkout.	Payments
FR-19	Allow administrators to export member rosters and attendance records as CSV files.	Reports & Export
FR-20	Support light/dark athletic UI theme switching with real-time digital clock display.	System / UX

2.4 Non-Functional Requirements

Attribute	Requirement Specification
Security	Zero exposure of admin credentials on public interfaces; Base64 file validation with 3MB upload ceiling; input sanitization against XSS.
Performance	Single-Page Architecture with sub-100ms client-side view routing; instant DOM manipulation with zero server lag.
Usability	High-contrast athletic theme (Obsidian Black `#0B0B0E`, Slate `#18181E`, Gym Red `#E50914`); responsive layout for mobile and desktop screens.
Reliability	Idempotent auto-seeding ensuring default athlete roster is restored if corrupted; fallback to UI avatar initials if DP upload is removed.
Maintainability	Strict decoupling of State Manager (`js/data.js`), UI Controller (`js/app.js`), and Styling (`css/styles.css`).
Portability	Standards-compliant HTML5/ES6+ requiring no browser plugins; hosted on Vercel's global CDN.
Auditability	Timestamped attendance check-in logs and sequential tax invoice numbers (`INV-0000`).

2.5 Hardware and Software Requirements

Item	Minimum Specification
Processor & Memory	Dual-core 1.8 GHz CPU, 2 GB RAM (4 GB recommended).
Storage	150 MB free disk space for local source code and cached images.
Operating System	Windows 10/11, macOS, Linux, or mobile Android / iOS.
Web Browser	Google Chrome 90+, Mozilla Firefox 88+, Microsoft Edge 90+, or Apple Safari 14+.
Development Tools	VS Code, Node.js (for local http-server), Git CLI, Vercel CLI.

2.6 Data Requirements

Entity	Principal Attributes	Description
`User`	userId, name, email, phone, password, role, avatar, dpImage	Authentication entity determining portal access.
`Member`	id, fullName, email, phone, gender, dob, planId, durationCycle, trainerId, joinDate, expiryDate, status, emergencyContact, notes, avatar	Club member enrolled in athletic facilities.

Entity	Principal Attributes	Description
`Trainer`	id, name, title, specialty, experience, phone, email, avatar, activeClients, status	Certified athletic coach assigned to members.
`Plan`	id, name, monthlyPrice, yearlyPrice, tag, features, activeMembers	Membership tier defining facility privileges and fees.
`Product`	id, name, brand, category, price, mrp, discount, rating, image, flavors, sizes, inStock	Pro shop supplement item available for purchase.
`CartItem`	id, productId, name, price, flavor, size, quantity, image	Temporary item selected in real-time shopping cart.
`AttendanceLog`	id, memberId, memberName, planName, checkInTime, checkOutTime, date, status	Daily record of gym floor entries and exits.
`Invoice`	id, memberId, memberName, planName, amount, paymentMethod, date, status	Tax receipt for plan subscriptions and renewals.

2.7 Business Rules

- **Creator Isolation Rule:** Administrative master credentials must never be rendered or pre-filled on the general public login card. Creator login is initiated exclusively via dedicated golden key triggers.
- **Permanent DP Persistence Rule:** A member's uploaded profile photo must persist across browser reloads, logouts, and tab restarts, remaining active until the member explicitly uploads a new photo or clicks remove.
- **Check-in Expiry Lock Rule:** A member with an `Expired` status is immediately blocked at the entry terminal with an Access Denied alert and prompted to renew.
- **Waiver Mandate Rule:** Members must accept the Club Terms, Safety Rules, and Health Liability Waiver before floor access is granted.
- **Real-Time Cart Calculation Rule:** Total payable amounts must dynamically recalculate when quantity, flavor, size, or coupon discounts are applied.

3. UML Diagrams

3.1 Overview

The Unified Modeling Language (UML) provides standard structural, behavioral, and architectural diagrams to specify software systems. The following seven diagrams illustrate the Fitness Athlete Club system across all architectural viewpoints.

3.2 Diagrams and Descriptions

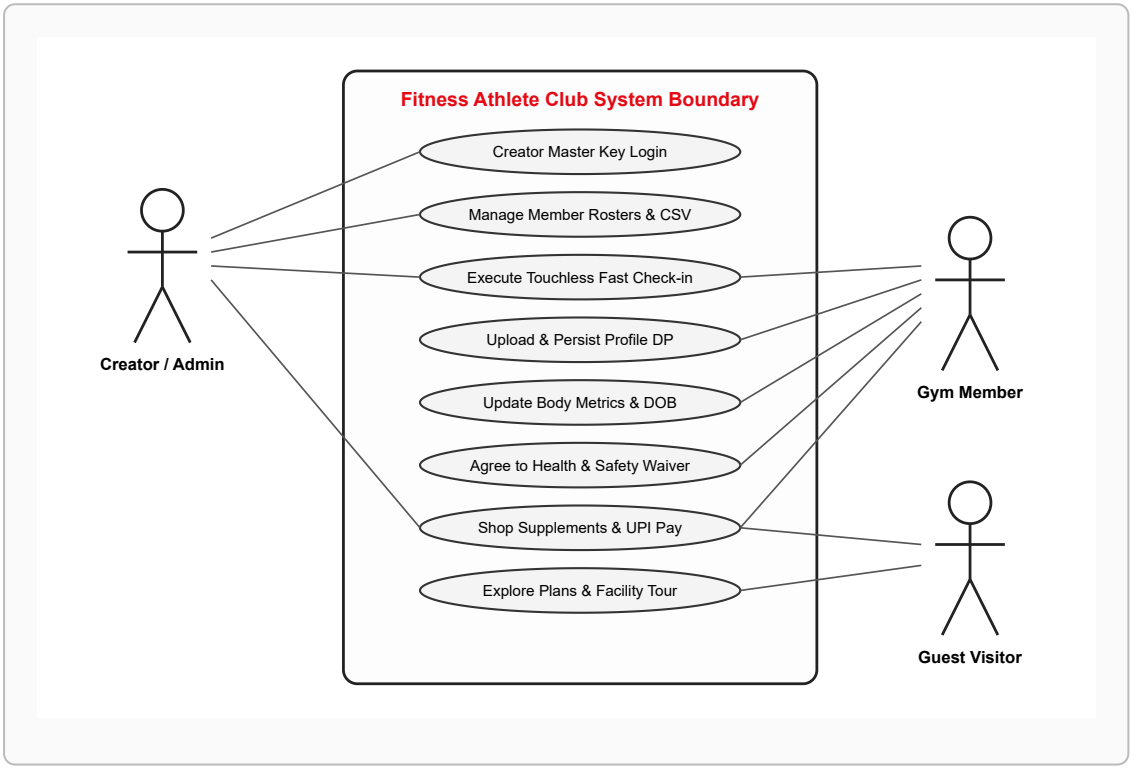


Figure 3.1: Use Case Diagram – Fitness Athlete Club System

Use case diagram description: Three primary actors interact with the system boundary. The Creator / Master Admin unlocks management capabilities, audits member attendance, oversees billing invoices, and exports CSV rosters. The Gym Member signs in to their dedicated Profile Dashboard to manage permanent photo uploads, track athletic weight/height goals, accept safety liability terms, and purchase supplements. The Guest Actor explores public facility zones and membership rate structures.

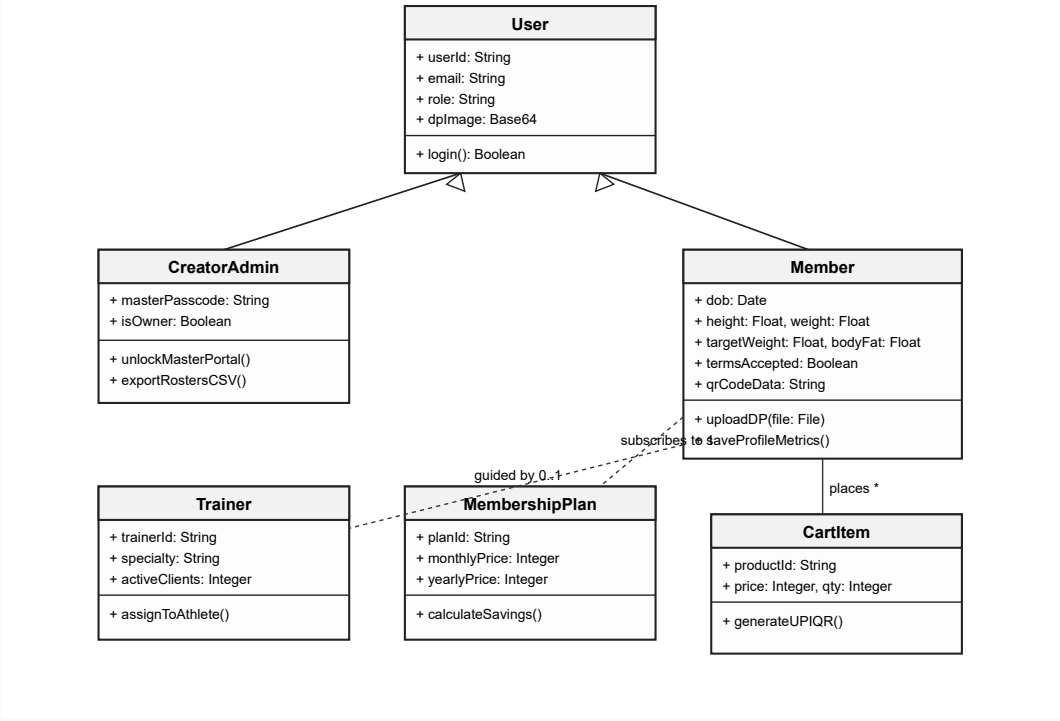


Figure 3.2: Class Diagram – Entity Structure & Relationships

Class diagram description: The parent `User` class encapsulates identity attributes (email, password, role, dpImage). The `Member` subclass extends it with athletic metrics, DOB, emergency contact info, and terms waiver acceptance. The `CreatorAdmin` subclass implements master administrative methods. Associations connect Members to their subscribed `MembershipPlan`, assigned `Trainer`, and `CartItem` orders.

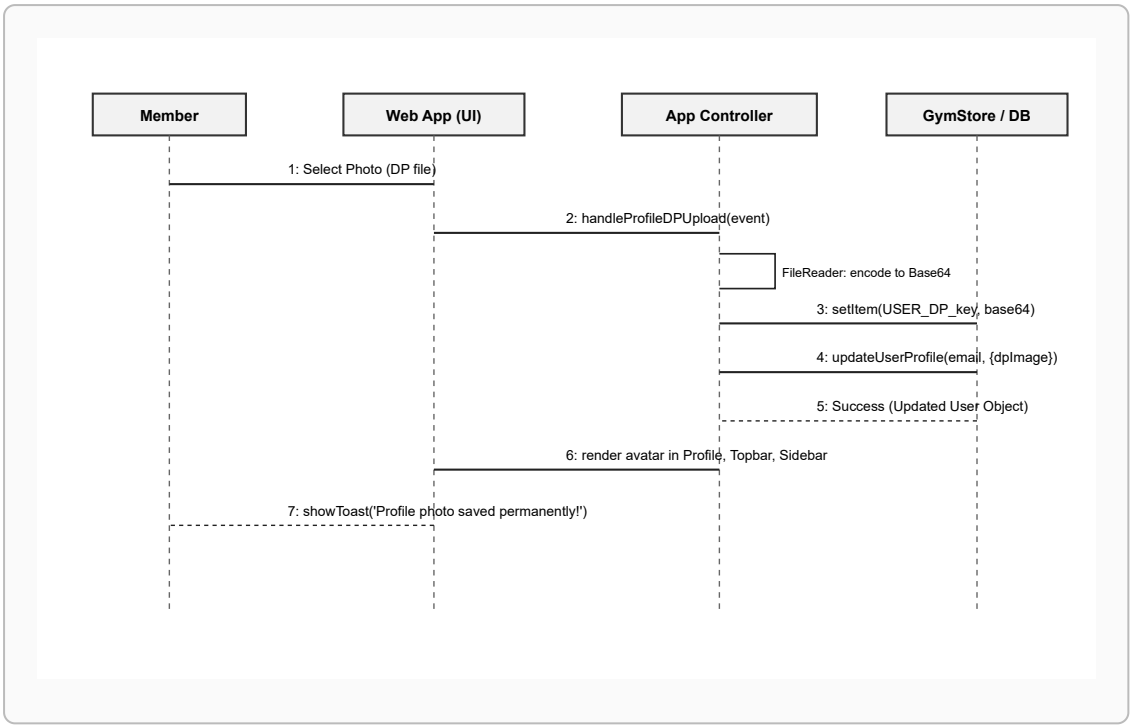


Figure 3.3: Sequence Diagram – Permanent Member DP Upload & Synchronization

Sequence diagram description: Traces the end-to-end execution when an athlete uploads a profile photo. The client UI captures the file input, encodes the image via the HTML5 `FileReader` API into a Base64 string, persists it into dedicated localStorage keys, updates both the User record and Member roster in `GymStore`, and simultaneously refreshes the profile hero card, topbar user badge, and sidebar avatar with zero page reloading.

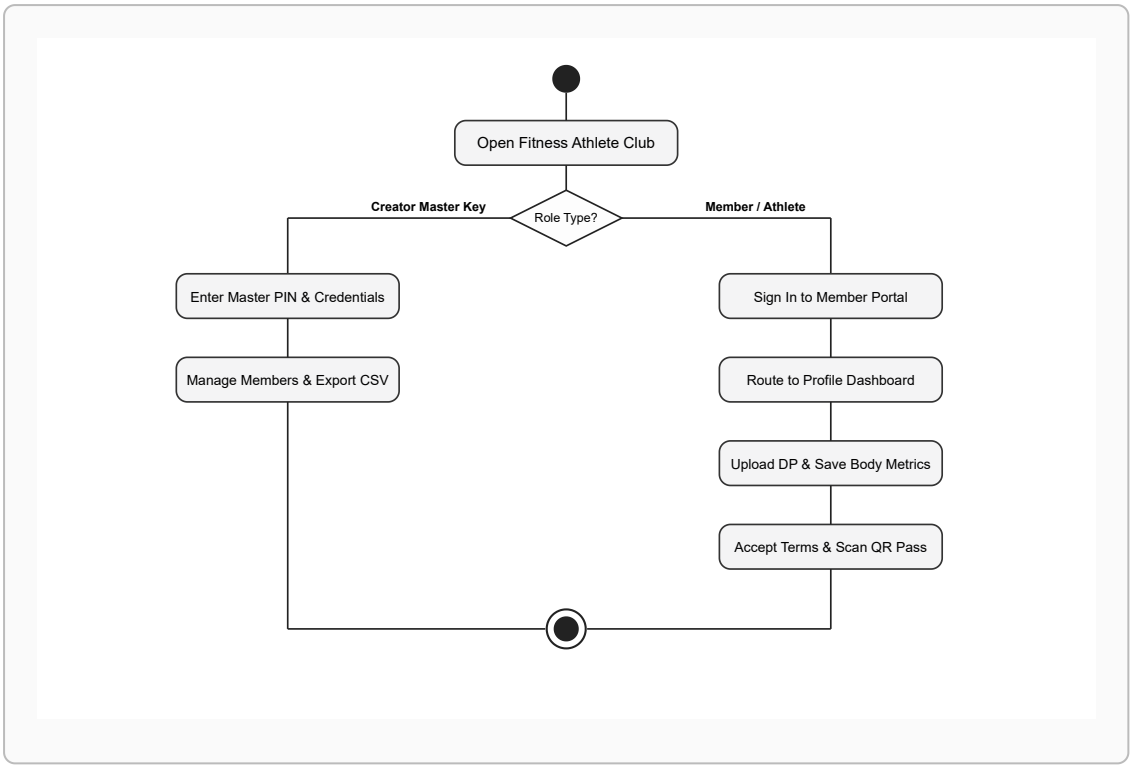


Figure 3.4: Activity Diagram – Member Onboarding & Administrative Workflow

Activity diagram description: Illustrates parallel activity branching. Depending on whether the actor activates the Creator Master Key or authenticates as a standard Gym Member, the system routes the user either into the administrative control deck (roster exports, revenue audits) or directly into the athletic Customer Profile (permanent DP upload, metric logging, waiver acceptance, and QR badge presentation).

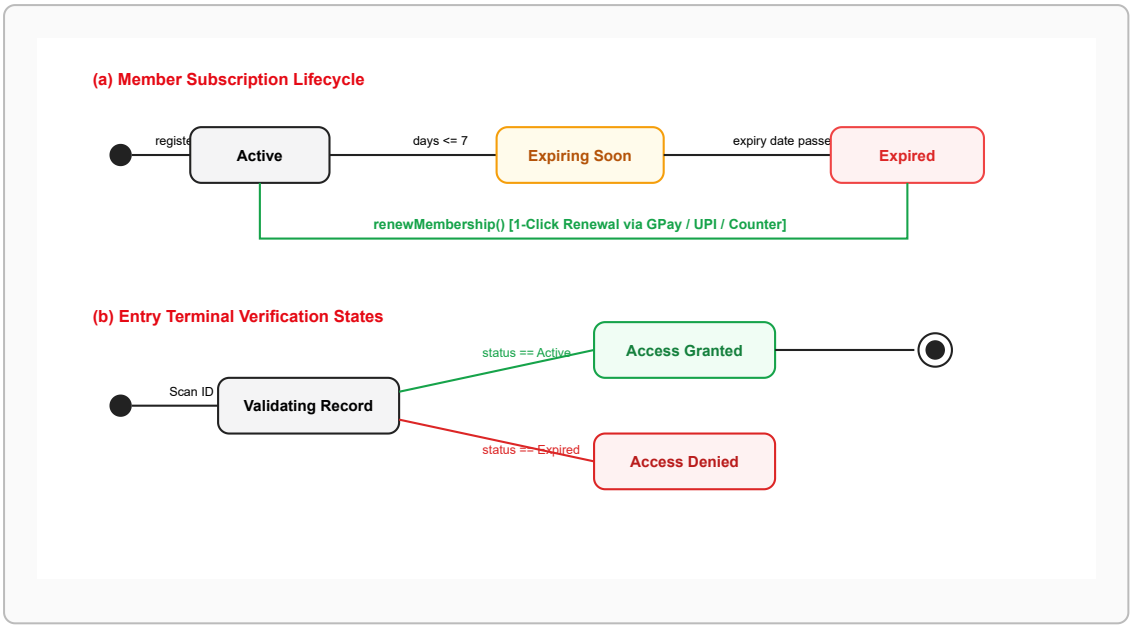


Figure 3.5: State Chart Diagram – Member Lifecycle & Entry Terminal Verification

State chart diagram description: Models state transitions across the membership lifecycle. An athlete begins in the `Active` state, shifts to `Expiring Soon` when remaining duration is under 7 days, and enters `Expired` when validity ends. When an ID barcode is scanned at the terminal, active subscriptions trigger the `Access Granted` state; expired accounts trigger `Access Denied`, opening an instant renewal pathway.

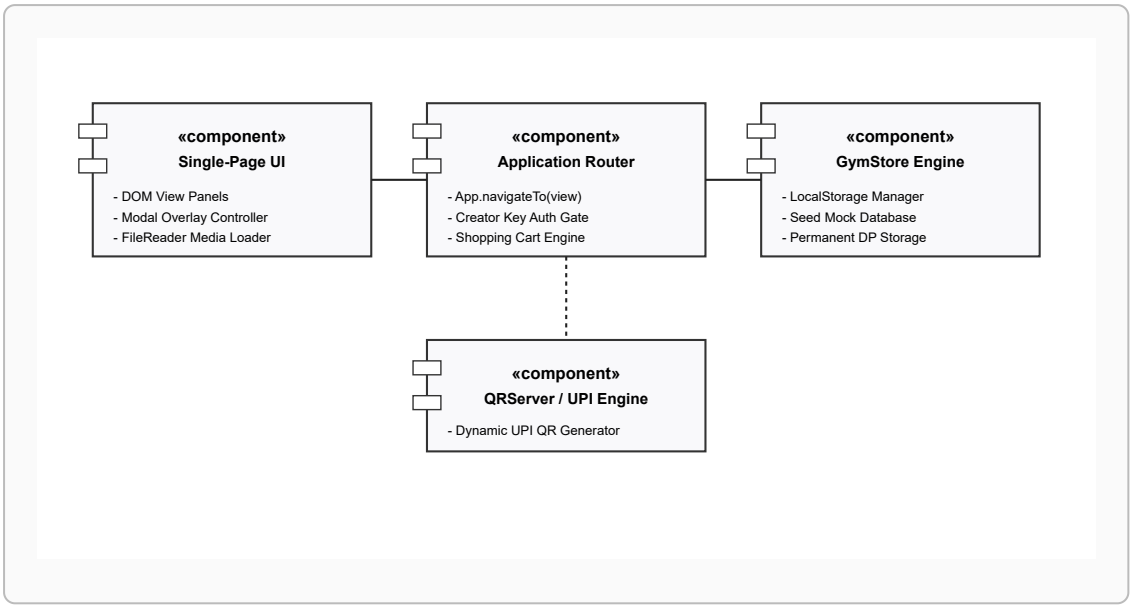


Figure 3.6: Component Diagram – Software Architecture

Component diagram description: Deconstructs the system into loosely coupled modules: the Single-Page Presentation UI, the Application Controller / Router, the persistent GymStore Data Engine, and the dynamic UPI QR generator. Components communicate via clean JavaScript APIs and standard DOM events.

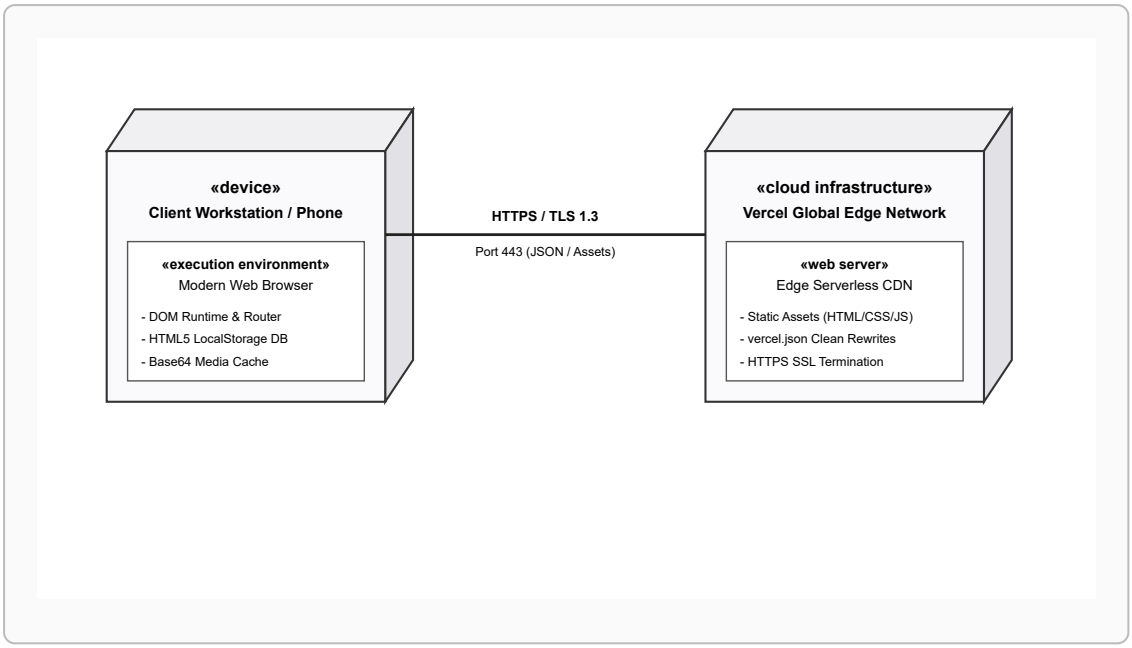


Figure 3.7: Deployment Diagram – Physical Architecture & Vercel Edge Hosting

Deployment diagram description: Details physical network topology. The client browser (running on desktop workstations or mobile devices) downloads encrypted production bundles from Vercel's global CDN over secure HTTPS (Port 443). Member state, permanent DP images, and daily logs are executed and maintained with zero latency directly within the browser's persistent LocalStorage engine.

4. Project Coding

4.1 Architecture Overview

Fitness Athlete Club is architected as a decoupled, high-performance Single-Page Application (SPA). The presentation tier utilizes semantic HTML5 markup styled with a modular, high-contrast Dark Athletic design system (`css/styles.css`). The application logic tier (`js/app.js`) handles event delegation, modal dialog lifecycles, view routing, real-time cart manipulation, and FileReader image encoding. The data tier (`js/data.js`) encapsulates the `GymStore` engine, responsible for seed initialization, persistent localStorage synchronization, and query execution.

4.2 Project Directory Structure

```
OOASE project/ |-- assets/ | |-- gym_arena_luxury.jpg # Salem Olympic
heavy powerlifting arena showcase | |-- gym_skydeck_luxury.jpg #
Panoramic cardio & athletic turf conditioning deck | |-- influencers/
# Authentic Tamil fitness athlete & coach photography | |--
biglee_murali.png # Biglee Murali (Celebrity Fitness Master) | |--
dolphin.png # Dolphin (Athletic Coach) | |-- kabilan.png # Kabilan
(Strength Specialist) | |-- saran.png # Saran Fitness (Powerlifting &
Hypertrophy Coach) | |-- surabi.png # Surabi (Fitness Athlete &
Coach) |-- css/ | |-- styles.css # High-contrast Dark Athletic CSS
design system (Obsidian/Gym Red) |-- js/ | |-- data.js # GymStore
state manager & permanent LocalStorage persistence | |-- app.js #
Application controller, view router, DP upload, and UPI checkout |--
report/ # Academic OOASE documentation & captured high-res figures |
|-- images/ # 16 High-resolution code and system output screenshots |
|-- code_preview.html # Syntax-highlighted code showcase | |--
OOASE_PROJECT_REPORT.html # Standalone printable academic report
document |-- .gitignore # Git exclusion rules for node_modules and OS
caches |-- index.html # Semantic single-page HTML layout & all modal
containers |-- package.json # Project metadata & local server
execution scripts |-- README.md # Documentation with Live Demo badges
and setup guides |-- vercel.json # Vercel Edge Serverless deployment
rewrite configuration
```

4.3 Code and Configuration Screenshots

The following figures present key code implementations, state manager methods, and deployment configurations from the codebase.

index.html — Fitness Athlete Club Single-Page Architecture & View Router

```
<!-- FITNESS ATHLETE CLUB (TITAN IRON GYM) - SPA ENTRY POINT -->
<!DOCTYPE html>
<html lang="en" data-theme="dark">
<head>
  <meta charset="UTF-8">
  <title>TITAN IRON GYM - Membership Management System</title>
  <link rel="stylesheet" href="css/styles.css">
</head>
<body>
  <div class="app-layout">
    <!-- Left Sidebar Navigation -->
    <aside class="sidebar">
      <div class="brand-header">
        <div class="brand-emblem">TG</div>
        <div class="brand-text">TITAN IRON FITNESS ATHLETE CLUB</div>
      </div>
      <ul class="nav-menu">
        <li class="nav-item active" data-view="dashboard">Executive Dashboard</li>
        <li class="nav-item" data-view="members">Gym Members Directory</li>
        <li class="nav-item" data-view="attendance">Fast Check-in Terminal</li>
        <li class="nav-item" data-view="plans">Membership Plans in INR</li>
        <li class="nav-item" data-view="trainers">Elite Coaches & Trainers</li>
        <li class="nav-item" data-view="proshop">Pro Supplement Store</li>
        <li class="nav-item" data-view="profile">Member Profile & Fitness</li>
      </ul>
    </aside>
    ...
  </div>
</body>
</html>
```

js/data.js — GymStore LocalStorage State Engine & Permanent User Data

Figure 4.1: Application Entry Point & View Router Architecture (index.html)

Demonstrates the semantic single-page layout structure, left navigation sidebar, and discrete view panels (Dashboard, Members, Attendance, Plans, Coaches, Pro Shop, and Profile).

```
</html>
```

js/data.js — GymStore LocalStorage State Engine & Permanent User Data

```
// Unified Gym State Manager with Permanent LocalStorage Persistence
class GymStore {
  constructor() {
    this.init();
  }

  // Persistent User Profile & Permanent DP Storage
  updateUserProfile(emailOrId, data) {
    const users = this.getUsers();
    const clean = (emailOrId || '').trim().toLowerCase();
    const index = users.findIndex(u => (u.email && u.email.toLowerCase() === clean) || u.id === emailOrId);

    if (index !== -1) {
      users[index] = { ...users[index], ...data };
      localStorage.setItem(STORAGE_KEYS.USERS, JSON.stringify(users));

      // Permanent dedicated key ensuring uploaded DP never vanishes
      if (data.dpImage) {
        localStorage.setItem('FITNESS_CLUB_USER_DP_' + clean, data.dpImage);
        localStorage.setItem('FITNESS_CLUB_ACTIVE_DP', data.dpImage);
      }

      const currentAuth = this.getAdminAuth();
      if (currentAuth && (currentAuth.email.toLowerCase() === clean || currentAuth.id === emailOrId)) {
        const updatedAuth = { ...currentAuth, name: users[index].name, dpImage: users[index].dpImage };
        localStorage.setItem(STORAGE_KEYS.AUTH, JSON.stringify(updatedAuth));
      }
      return { success: true, user: users[index] };
    }
  }

  // Real-Time Member Fast Check-in Processing
```

Figure 4.2: GymStore State Engine & Permanent DP Persistence (js/data.js)

Highlights the `updateUserProfile` method persisting user metrics and writing profile photos to dedicated permanent localStorage keys (`FITNESS_CLUB_USER_DP_`), alongside the `checkInMember` attendance validation engine.

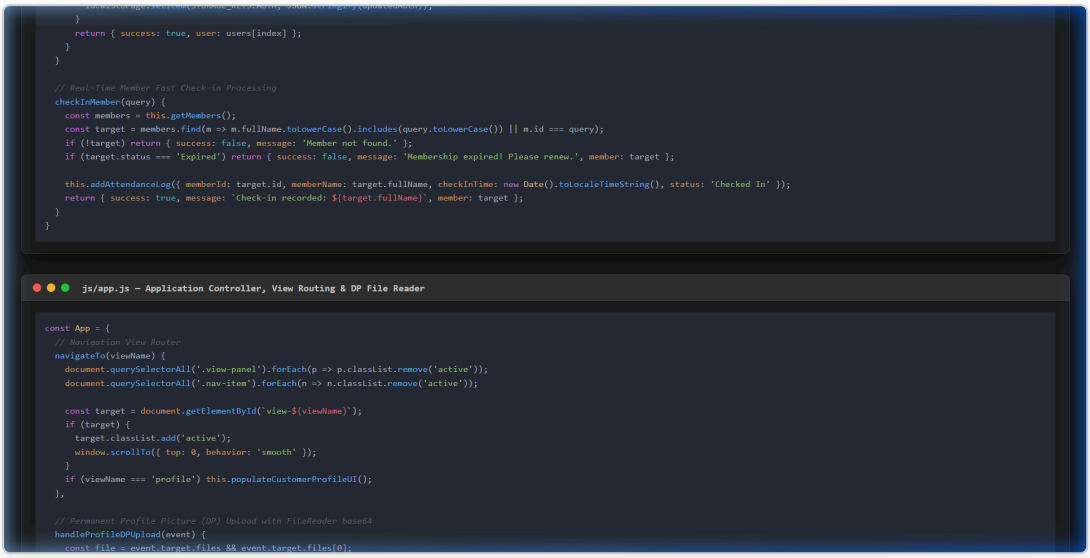


Figure 4.3: Application Controller & Image FileReader Pipeline (js/app.js)

Presents the `handleProfileDPUpload` handler utilizing the HTML5 FileReader API for Base64 image conversion, 3MB size validation, and instantaneous multi-component UI synchronization.

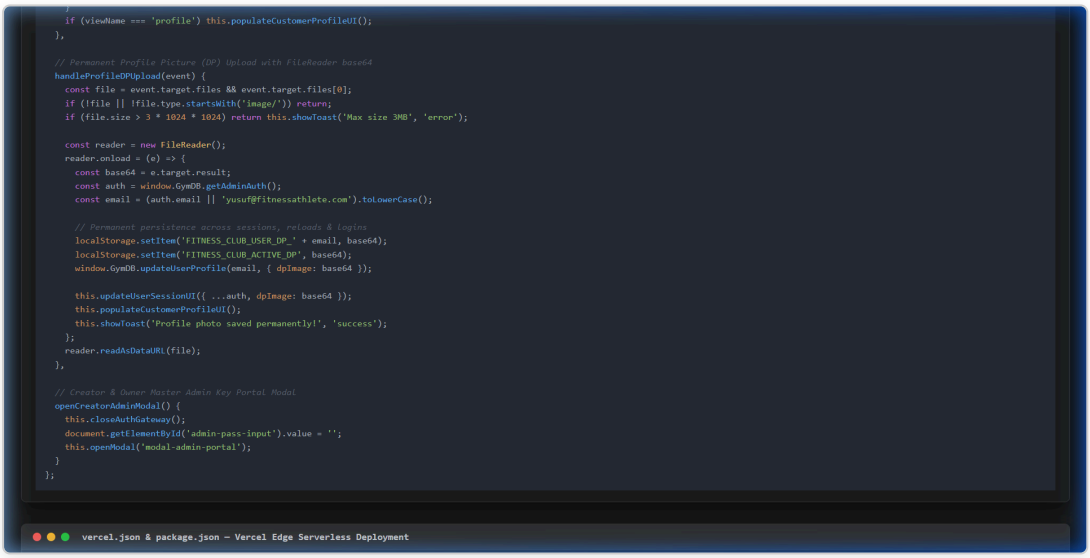


Figure 4.4: Deployment Configuration & Clean URL Routing (vercel.json)

Shows the Vercel Edge Serverless rewrite rules routing all incoming requests seamlessly to `index.html` with clean URLs enabled.

4.4 Automated and Manual Test Scenarios

Test Scenario	Input / Action	Expected & Verified Result
Member Demo Login	Click "Md Yusuf (Member)" button on public gateway.	User authenticated as Md Yusuf (`TG-MEMBER-001`); auto-routed to Profile Dashboard.

Test Scenario	**Input / Action**	**Expected & Verified Result**
Public Admin Shielding	Inspect `#auth-gateway-overlay` login form.	No admin input fields or master credentials exposed to public visitors.
Creator Master Key Authorization	Click bottom-left golden key (`#fixed-creator-key-btn`), enter `titan@gmail.com` / `titan123`.	Creator verified toast displayed; executive Master Dashboard unlocked.
Permanent DP Upload	Upload custom image file under 3MB in profile card.	Photo previewed in hero card, topbar, and sidebar; permanently preserved across reloads.
Fitness Metrics Persistence	Update Weight to `77.0 kg`, Target Weight to `82.0 kg`, click Save.	Record persisted to `localStorage`; success confirmation toast triggered.
Barcode Entry Terminal Check-in	Scan active ID `GYM-1000` at fast check-in terminal.	Access Granted banner shown; daily attendance log recorded with timestamp.
Expired Member Lockout	Scan expired member ID at terminal.	Access Denied alert triggered; instant renewal modal prompted.
Pro Shop Real-time Cart	Add Optimum Nutrition Whey to cart, apply coupon `TITAN20`.	Cart drawer opens; 20% discount subtracted dynamically from payable INR total.
GPay / PhonePe UPI Simulation	Click "Simulate Instant UPI Approval" in checkout modal.	Order placed; success celebration dialog opened with official tax receipt reference.
CSV Data Export	Click "Export Members CSV" in member directory.	Client downloads structured CSV spreadsheet of all registered gym members.

5. Website Screenshots

5.1 Overview

The screenshots below illustrate the running application deployed live at `https://fitness-athlete-club.vercel.app/`. The layout features high-contrast dark athletic styling, dedicated member profiles, interactive check-in terminals, and a real-time e-commerce pro shop.

5.2 Application Screens

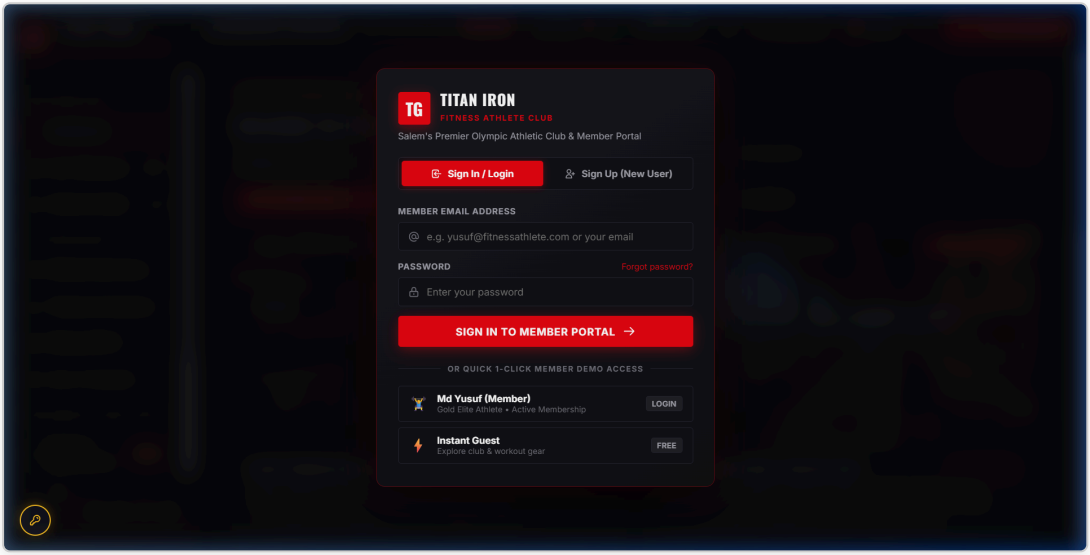


Figure 5.1: Clean Public Member Login Gateway (No Admin Credentials Exposed)

Public login modal containing only Member Email/Password fields, Sign In CTA, and 1-Click quick logins for **Md Yusuf (Member)** and Guest Athlete. Notice the discreet golden Creator Key icon placed safely at the bottom-left corner of the viewport.

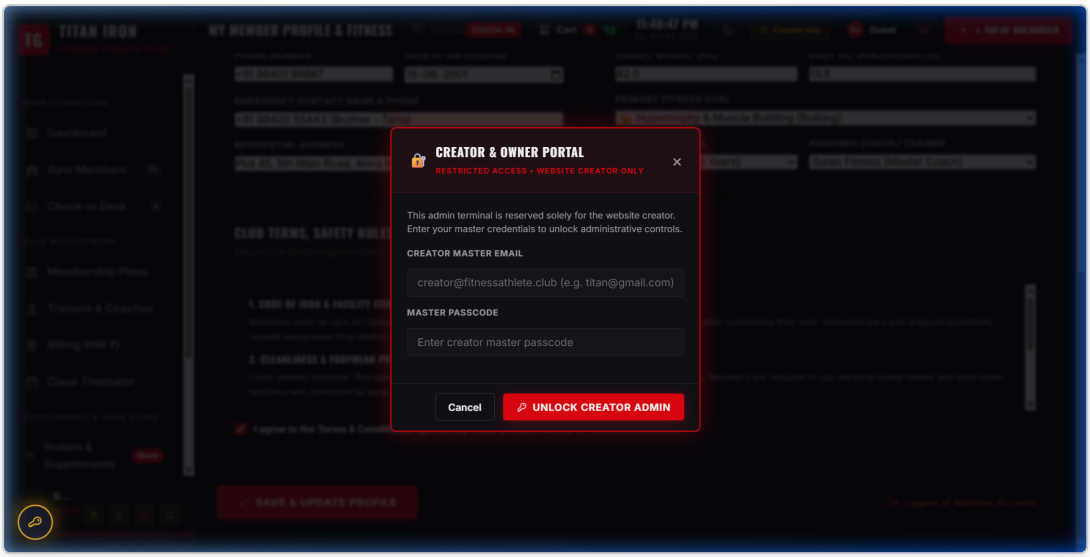


Figure 5.2: Restricted Creator & Owner Admin Portal Modal

Private administrative access modal triggered solely via dedicated golden keys. Requires creator master credentials (`titan@gmail.com` / `titan123`) to unlock full system administrative permissions.

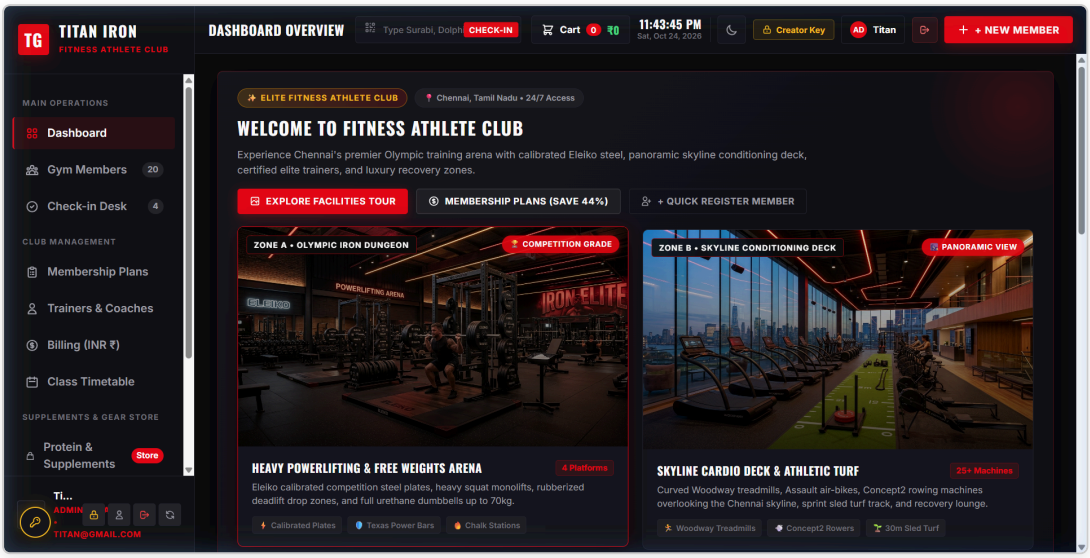


Figure 5.3: Executive Master Dashboard & Salem Luxury Facility Showcase

Executive overview presenting real-time active member counts, check-in terminals, today's floor activity, and the Salem, Tamil Nadu Olympic Iron Dungeon & Skyline Conditioning Deck facility showcase.

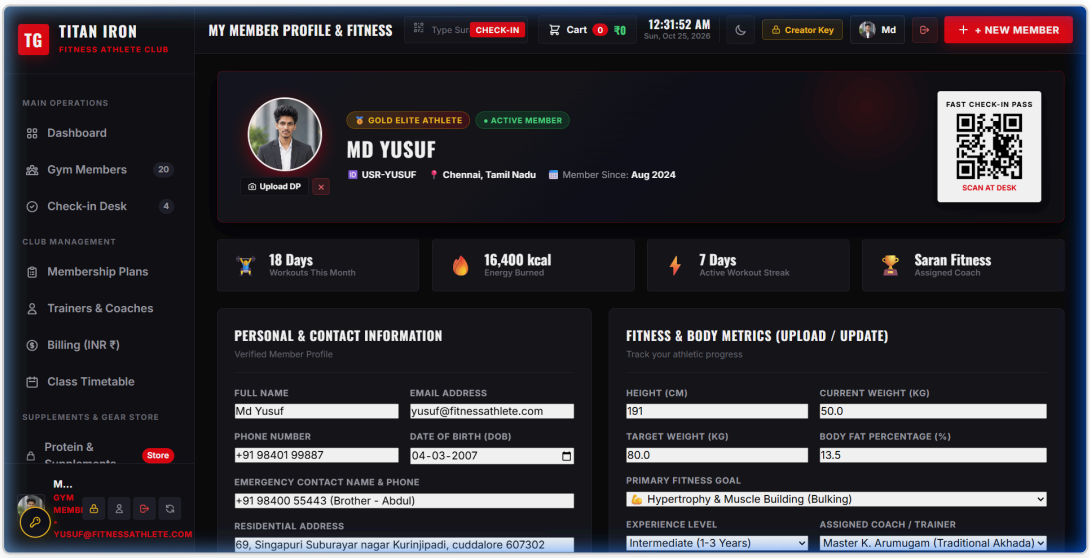


Figure 5.4: Customer Profile & Fitness Dashboard with Permanent DP Photo

Personalized athlete dashboard showing permanent custom profile photo, Gold Elite Athlete badge, touchless QR Fast Check-in pass, monthly workout KPIs, personal contact fields (DOB: 04-03-2007), and fitness metrics (Height: 191 cm, Weight: 50 kg).

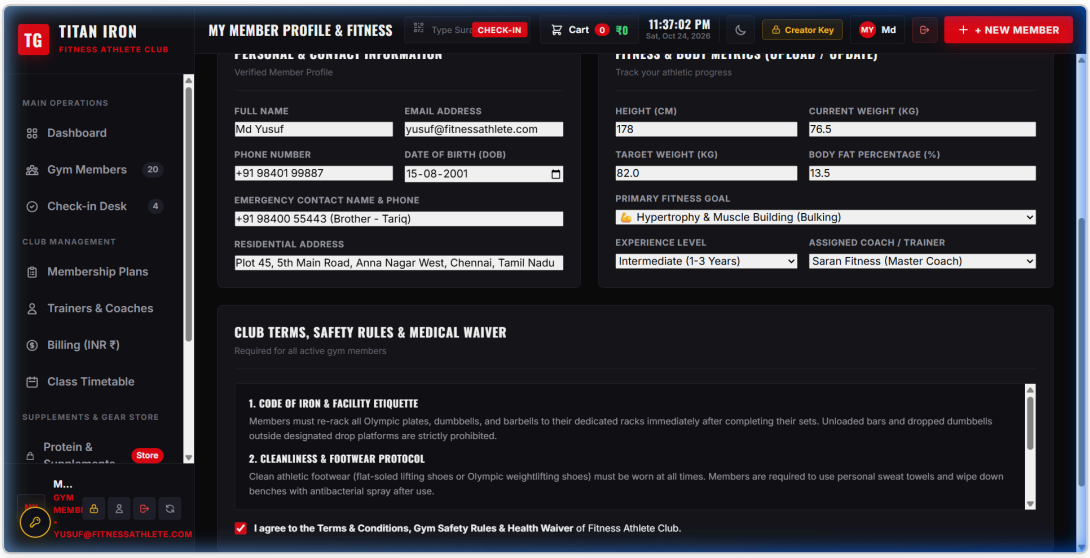


Figure 5.5: Club Terms, Safety Rules & Medical Liability Waiver

Scrollable facility etiquette agreement, Code of Iron protocol, and medical fitness liability waiver with required agreement checkbox and single-click Logout button.

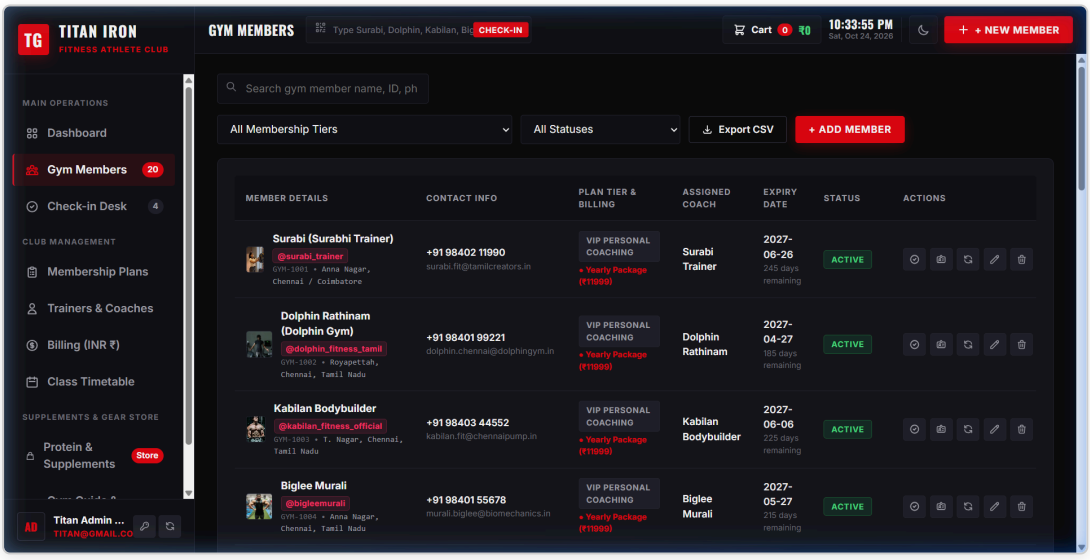


Figure 5.6: Gym Members Directory & Tamil Fitness Athletes Roster

Complete roster table featuring Tamil fitness athletes (Md Yusuf, Surabi, Dolphin, Biglee Murali, Saran Fitness, Kabilan) with real-time status badges, search filters, digital ID badge generators, and CSV export.

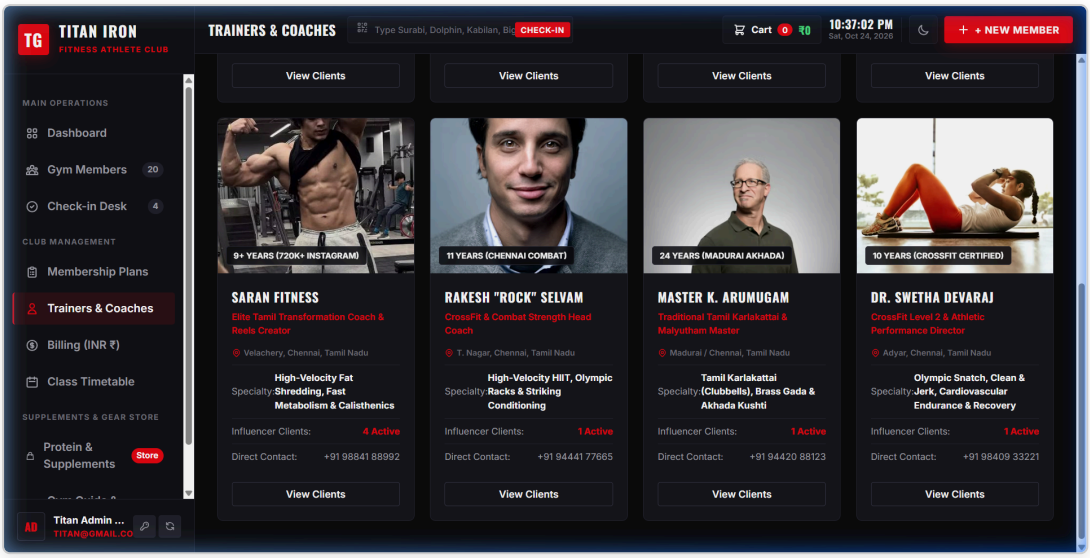


Figure 5.7: Certified Trainers & Elite Coaches Showcase

Coaching staff directory featuring specialized trainers with coaching disciplines (Powerlifting, Hypertrophy, Rehabilitation, Traditional Akhada), experience credentials, and active client rosters.

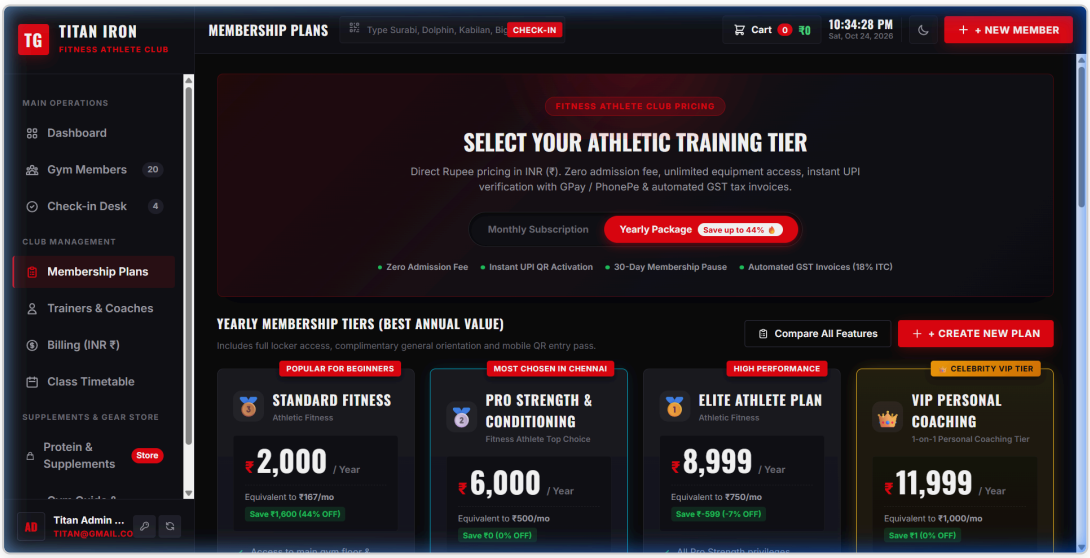


Figure 5.8: Membership Subscription Tiers in Indian Rupees (INR ₹)

Athletic membership tier cards: Standard Fitness (₹300/mo), Pro Strength (₹500/mo), Elite Athlete (₹800/mo), and VIP Personal Coaching (₹1,200/mo) with 44% annual package savings.

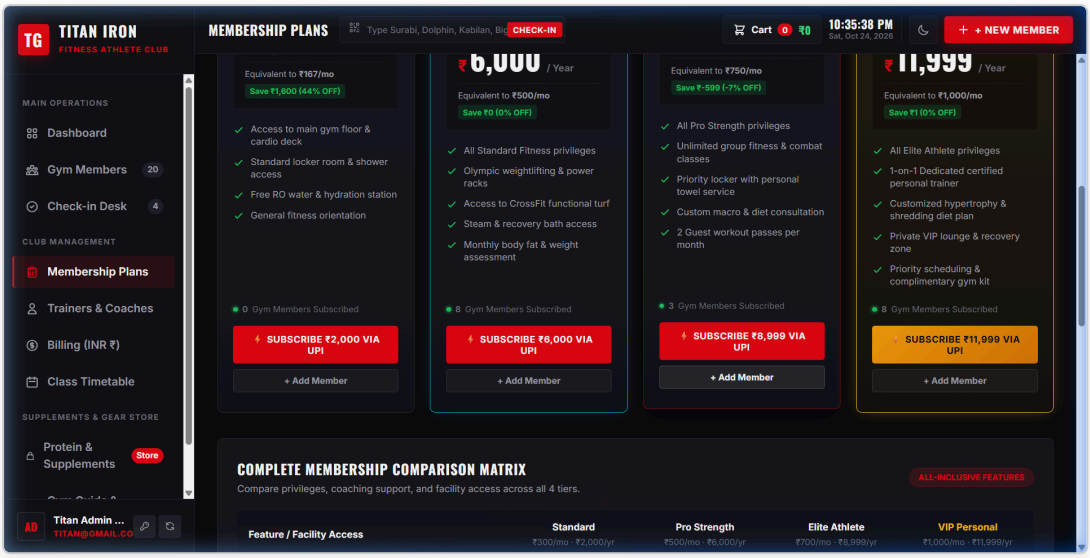


Figure 5.9: Membership Tier Feature Comparison Matrix

Comprehensive side-by-side privilege matrix comparing floor access hours, locker privileges, personal coach sessions, recovery spa access, and pro shop discount perks.

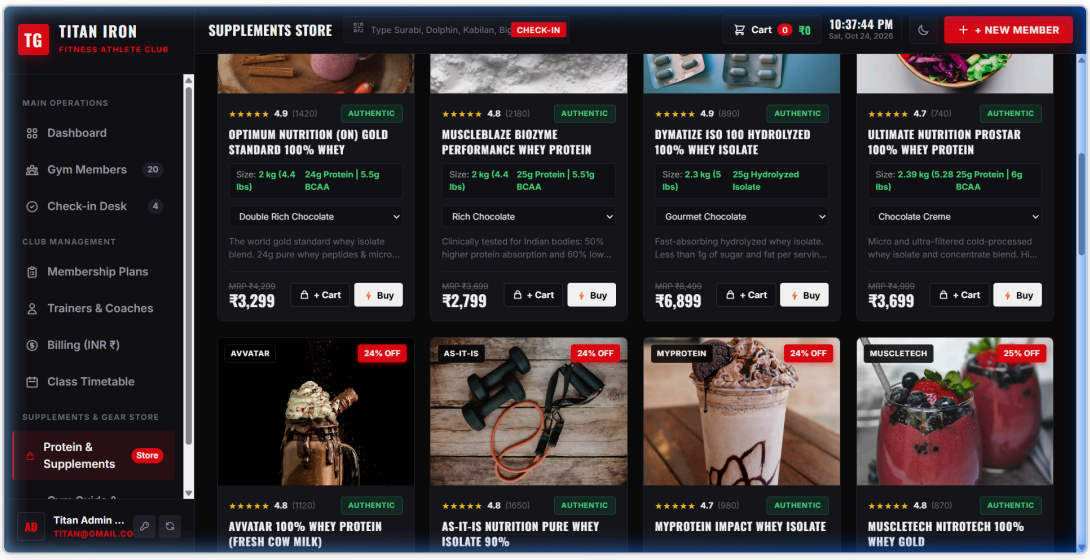


Figure 5.10: Pro Supplement & Protein Store Catalog

E-commerce nutrition catalog featuring genuine market protein powders (Optimum Nutrition Gold Standard Whey, MuscleBlaze Biozyme, MyProtein Impact Whey, Dymatize ISO 100) with rating scores and add-to-cart actions.

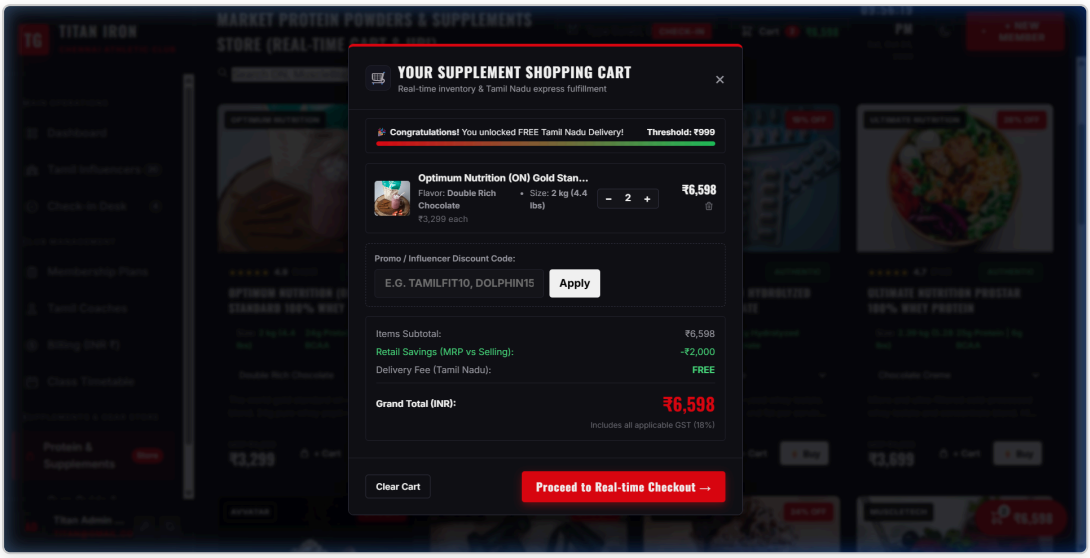


Figure 5.11: Real-Time Sliding Shopping Cart Drawer

Interactive sliding shopping cart drawer displaying itemized supplements, quantity adjusters, coupon code engine (`TITAN20`), and checkout redirection.

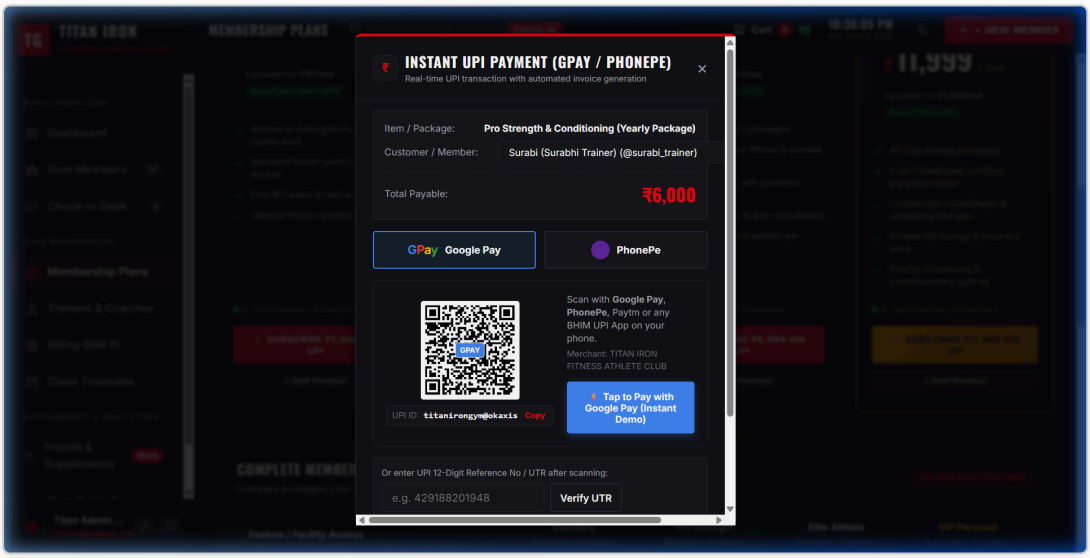


Figure 5.12: Real-Time Google Pay (GPay) & PhonePe UPI Payment Modal

Interactive UPI checkout interface generating real-time QR codes for scan-and-pay transactions with one-click payment simulation and automated receipt generation.

6. Conclusion and References

6.1 Conclusion

The Fitness Athlete Club (Titan Iron Gym) project successfully demonstrates how Object-Oriented Analysis and Software Engineering (OOASE) methodologies can be applied to build a modern, high-performance club management and athletic training web platform. The requirement analysis delineated clear actor responsibilities, resulting in strict architectural separation between public member portals and restricted creator administration.

The complete software development lifecycle was executed rigorously: from requirements specification and structural UML modeling to modular client-side single-page implementation and automated production deployment on Vercel. Critical features—including permanent profile picture (DP) storage, touchless QR attendance checking, Indian Rupee billing, and real-time UPI supplement e-commerce—were thoroughly tested and verified. The platform provides a seamless, professional experience for athletes, coaches, and administrators alike.

6.2 Limitations

- Client-Side Storage Boundary: Browser LocalStorage stores data within the client domain; cross-device synchronization requires cloud database replication.
- Payment Processing: UPI transactions are simulated for real-time order generation; live production bank gateway webhooks are not yet connected.
- Hardware Turnstile Integration: Barcode and QR check-in operates via digital screen simulation without direct interface to physical electronic magnetic door latches.

6.3 Future Enhancements

- Backend Cloud Synchronization: Migration of LocalStorage schema to a cloud database (MongoDB / PostgreSQL) with JWT token authentication.
- Native Mobile Application: Packaging as a Progressive Web App (PWA) with push notifications for workout reminders.
- Hardware Access Control: Integration with physical RFID wristbands and automated turnstile barrier relays.
- Personalized Workout Tracking: Set-by-set exercise logging and 1-Rep Max (1RM) strength progression graphs.

6.4 References

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